Consciousness and IIT

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Overview

What is Integrated Information Theory (IIT)?

- **Origin**: Developed by neuroscientist Giulio Tononi.
- **Fundamental Approach**: Unlike other theories that start from the brain (physical) and try to explain consciousness, IIT starts from **phenomenology** (experience) and derives physical requirements.
- **Key Metric (Φ)**:
  - Φ (Phi) represents the quantity of consciousness.
  - A system is conscious if and only if it has $\Phi > 0$.
  - The higher the Φ, the more conscious the system is.

The Five Axioms of Phenomenology

IIT posits that every conscious experience satisfies five essential properties (axioms):

1. **Intrinsic Existence**: Consciousness exists for the subject; it is real.
2. **Composition**: Experience is structured and composed of parts (e.g., a blue book on a table).
3. **Information**: Experience is specific; it is *this* one particular experience out of purely distinct possibilities.
4. **Integration**: Experience is unified. You cannot experience the left visual field independently of the right visual field.
5. **Exclusion**: Experience is definite in content and spatio-temporal grain (it flows at a specific speed and has specific boundaries).

Irreducibility and the Minimum Information Partition

The Weakest Link Principle

Integration (Φ) is determined by the system's "weakest link". We must find the cut that causes the *least* loss of intrinsic cause-effect power.

- **Irreducibility**: A system is irreducible if it cannot be partitioned into independent parts without losing information.
- **Minimum Information Partition (MIP)**:
  - We test all possible ways to cut the system into parts.
  - The MIP is the specific cut that minimizes the difference between the whole and the partitioned system.
- **The Value of Φ**:

$$\Phi = \text{Distance}(\text{Whole} \parallel \text{MIP})$$

- Φ quantifies how much the whole is more than the sum of its parts, measured strictly across this weakest link.

Estimate Φ of Humans using EEG data

Validation

IIT can only be validated by measuring Φ of systems we know are conscious. That is, humans.

- We will measure and compare the integrated information between conscious and sedated individuals.
- We use a 64-channel EEG dataset with Awake and propofol sedated data
- We approximate Φ rather a proper calculation.

EEG Overview

- Electroencephalography (EEG) measures electrical activity generated by neural populations in the brain.
- EEG signals can be decomposed into **five major spectral bands**, each associated with functional states:
  - **Delta (0.5–4 Hz):** Deep sleep and unconscious states.
  - **Theta (4–8 Hz):** Drowsiness, meditation, memory encoding.
  - **Alpha (8–12 Hz):** Relaxed wakefulness, eyes-closed resting state.
  - **Beta (12–30 Hz):** Active thinking, focus, sensorimotor activity.
  - **Gamma (30–100 Hz):** High-level cognition, binding, consciousness-related processes.
- These spectral decompositions help interpret neural dynamics under awake vs. sedated conditions.

Method (1/2)

Estimating Mutual Information

- Discretize continuous EEG signals into bins.
- Use binned values to form empirical probability distributions.
- Compute entropy:

$$H(X) = -\sum_i p(x_i) \log p(x_i)$$

  and similarly for $H(Y)$ and joint entropy $H(X, Y)$.
- Mutual information:

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

Method (2/2)

Approximating Φ via Channel Subsampling

- Randomly sample 16 of the 64 EEG channels.
- Repeat this process 50 times with different subsets.
- For each subset, compute integration over 30-second EEG epochs.
- Average the 50 integration values to approximate Φ.

Results

Subject	Eyes Open	Eyes Closed	Sedation
sub-1010	4.8068	4.4755	4.0462
sub-1017	5.5163	6.2293	3.3788
sub-1022	6.2649	6.4356	6.2582
sub-1024	6.7651	6.8023	6.1351
sub-1033	5.4524	5.5421	2.5320
sub-1045	5.2325	5.2949	6.4032
sub-1046	6.4259	6.6649	6.1872
sub-1054	5.8539	6.6553	6.2689
sub-1055	5.9403	5.8223	4.1098
sub-1057	5.8275	6.6824	6.4885
sub-1060	5.2375	6.2199	5.2055
sub-1062	5.0487	4.8154	5.0291
sub-1064	6.4105	5.6766	5.2296
sub-1067	5.0232	5.8184	5.4621
sub-1068	5.4022	5.7414	4.4423
sub-1071	5.9287	6.1050	3.4828